Cosmic Ray Subsystem Report
On The Voyager-2 Flight Data System Anomaly
And The Subsequent Data Impact

Anomaly

On April 26, 2010 the Voyager Cosmic Ray Subsystem (CRS) team at the Goddard Space Flight Center (GSFC) was informed by the Jet Propulsion Laboratory (JPL) that the Deep Space Network (DSN) was unable to frame-sync the Voyager-2 telemetry starting on April 22 (DOY 112).

An investigation by JPL revealed that the onboard Flight Data System (FDS) computer, which formats the data to send back to Earth, suffered a flipped bit in a single memory location. Due to the almost 13 hour One Way Light Time (OWLT) the investigation of the anomaly and the solution took nearly a month to resolve. Thus, there was no Voyager-2 data for the time period from April 22 to May 21 (DOY 112 – 142). See **Appendix A** for the detailed emails and web site postings covering the anomaly investigation.

DOY 143

The first Voyager-2 data following the anomaly was a daily Experimental Data Record (EDR) for DOY 143. This file, however, would not process using the CRS team's standard Encyclopedia Generation software (*ENCGEN*). An investigation by Dr. Nand Lal (GSFC) revealed that the EDR contained a time discrepancy of about 2 hours and 28 minutes between the Earth Receive Time (ERT) and the Space Craft Event Time (SCET) over the expected OWLT difference. JPL determined that during the anomaly the onboard FDS clock did not increment resulting in the timing offset. A time check subroutine in the CRS *ENCGEN* software flagged each record due to this time discrepancy resulting in no data output and the failed processing.

To get around this problem Dr. Nand Lal (GSFC) created a specialized version of the production software that excluded the time check subroutine. This specialized software version was used to process Voyager-2 EDRs from DOY 143 – 175 (May 22 – June 24, 2010). Thus, the final Voyager-2 CRS EDRs processed with the specialized, 'no timecheck' software are: `crsf10143.b, crsf10147.b, crsf10153.b, crsf10161.b` and `crsf10168.b.` See **Appendix B** for the detailed emails and notes covering the time discrepancy between the ERT and SCET.

DOY 176 and 183

Additional complications arose starting with DOY 176. CRS was unable to process the Voyager-2 daily EDR for DOY 176 using the specialized software. The *ENCGEN* runtime output indicated that the entire EDR consisted of "`Overlapping Data`" and thus no output was generated. After an inquiry, JPL informed CRS that the most of the data was corrupted due to rain.

A subsequent investigation using EDR's for DOY 170 – 189 revealed, unexpectedly, that the DOY 176 EDR would process using the standard production software. The same investigation also revealed that significantly more data was processed from DOY 183 EDR with the standard production software than with the specialized software. With the specialized *ENCGEN* most of the DOY 183 records were also flagged as "`Overlapping Data`". This investigation was initiated as a data check against the final EDR: `crsf10175.b`. Because acceptable output was generated for DOY 176 and 183 using the standard *ENCGEN* software this production discrepancy between the two software versions was not investigated further.

FDS Correction

To correct the FDS time discrepancy on Voyager-2 JPL input a FDS SCLK reset on DOY 179:

2010-179/16:25:57.564 (ERT) = 2010-179/03:42:57.465 (SCET)

However, JPL informed CRS that the central data system would be unable to put the new correlation file in place until the DSS pass was over for DOY 181. As a result, data for DOY 180 and 181 would still have the old clock issue until the new file is on the system starting on DOY 182.

Daily EDRs for DOY 170 – 189 and the final EDR CRSF10176.b

Following the FDS correction the processing of the Voyager-2 final EDR `crsf10175.b` (DOY 176-183), which contains data from before and after the reset, failed using the specialized 'no timecheck' version of *ENCGEN*. The *ENCGEN* runtime output indicated that the entire EDR consisted of "`Overlapping Data`" and thus no output was generated. This final EDR was processable using our standard *ENCGEN* software, however, part of DOY 179 and all of DOY 180 were excluded (see yellow sections in Figure 1) due to time check flags:

"`FDSC inconsistent with earth received time`."

These exclusions directly followed the FDS SCLK reset (dashed line in Fig. 1).

As mentioned above, CRS reprocessed the daily EDRs for DOY 170-189 using both the standard *ENCGEN* software and the specialized 'no timecheck' version as a data check against the final EDR: `crsf10175.b`. This investigation revealed the following:

170	The standard *ENCGEN* software produced a slightly larger encyclopedia file: Standard Volume 1173263-1173350, Specialized Volume: 1173263-1173349
171 – 175	The standard and specialized *ENCGEN* software produced the same output
176	Will only process with the standard *ENCGEN* software. Discussed above.
177	The standard *ENCGEN* software produced a slightly larger encyclopedia file: Standard Volume 1173968-1174030, Specialized Volume: 1173968-1174029
178	The standard and specialized *ENCGEN* software produced the same output

179 The specialized *ENCGEN* software produced a larger encyclopedia file:
Specialized Volume 1174174-1174224, Standard Volume: 1174174-1174189
This will be discussed further below.

180 Will only process with the specialized *ENCGEN* software. This will be discussed further below.

181 –182 The standard and specialized *ENCGEN* software produced the same output

183 The standard *ENCGEN* software produced a larger encyclopedia file:
Standard Volume 1174511-1174606, Specialized Volume: 1174511-1174512

184 – 189 The standard and specialized *ENCGEN* software produced the same output

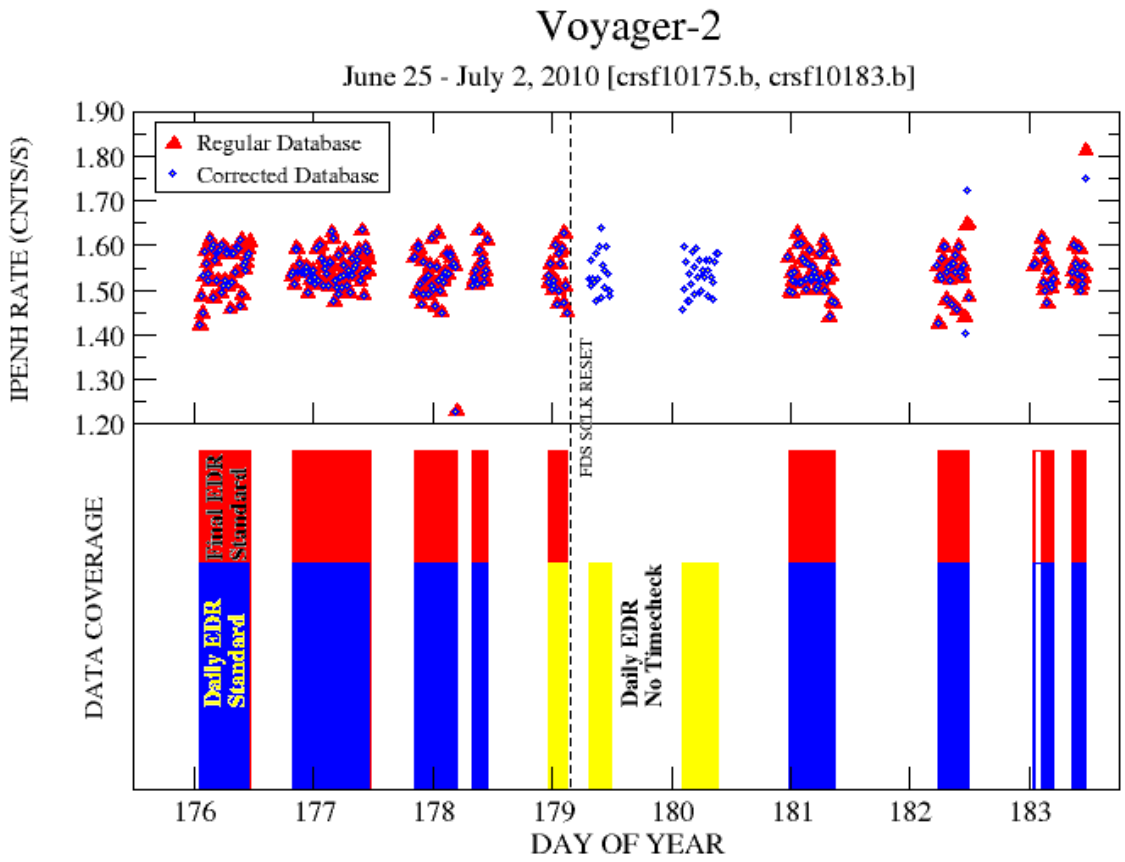


Figure 1 Top: Voyager-2 IPENH rate showing the data differences between the corrected and uncorrected data base. Bottom : Voyager-2 coverage representation . The red boxes represents the final EDR processed with the standard *ENCGEN* program. The blue represents the daily EDRs processed with the standard *ENCGEN* program. The yellow represents the daily EDRs processed with the specialized *ENCGEN* program.

The final Voyager-2 EDR `crsf10176.b` starts with DOY 176 and ends with DOY 183. It is suspected that the final EDR will not process with the specialized *ENCGEN* program because these two daily EDRs will not process with this version. This processing problem was not investigated further.

Database Correction for Final EDR CRSF10176.b

Since the encyclopedia file generated from the final Voyager-2 EDR `crsf10176.b` with the standard *ENCGEN* software is missing the last part of DOY 179 and all of DOY 180 we can only include it in the encyclopedia database if we can fill in the missing data. However, the encyclopedia database files are sorted by the analysis software using the starting volume number. Simply copying the two missing encyclopedia files generated from the daily EDRs into the database will be incorrect as the ending volume numbers are expected to be sequential as well. For example:

`crsf10175.b`	1173894 – 1174580	Final EDR
`crs10179.b`	1174174 – 1174224	Daily EDR
`crs10180.b`	1174282 – 1174310	Daily EDR

The easiest solution to fix the missing data is to replace the final encyclopedia file in the database with the daily encyclopedia files processed as noted.

`crsf10168.b`	1173141 – 1173813	no timecheck	Existing Final Encyclopedia File
`crs10176.b`	1173894 - 1173933	standard	Daily EDR
`crs10177.b`	1173968 - 1174030	standard	Daily EDR
`crs10178.b`	1174066 - 1174125	standard	Daily EDR
`crs10179.b`	1174174 - 1174224	no timecheck	Daily EDR
`crs10180.b`	1174282 - 1174310	no timecheck	Daily EDR
`crs10181.b`	1174367 - 1174404	standard	Daily EDR
`crs10182.b`	1174487 - 1174509	standard	Daily EDR
`crs10183.b`	1174511 - 1174606	standard	Daily EDR
`crsf10183.b`	1174595 - 1175172	standard	Existing Final Encyclopedia File

Although there is volume number overlap between the daily EDR for DOY 183 and the final EDR, `crsf10183.b`, the sort order is correct and the analysis programs will exclude the duplicate data. The time coverage of these daily data sets are represented by the blue and yellow rectangles in the bottom panel of Figure 1. The standard production version of *ENCGEN* was reinstated as the default software following this investigation:

"`/cosmicra/voyager/owlt/convertEDRtoVB`"
"`/cosmicra/voyager/encgen/src.test/encgen`"

See **Appendix C** for the detailed emails and notes covering the DOY 176-183 issues.

Second FDS Correction

JPL performed a second Voyager 2 FDS SCLK Reset on 2010-216/15:42 ERT. The purpose of this reset was to have the CCS clock and the FDS clock on the Voyager 2 to sync correctly in order to do sequences in the future.

Daily EDRs for DOY 216 - 217 and the final EDRs CRSF10210.b and CRSF10217.b

Following the second FDS correction the processing of the Voyager-2 daily EDR `crs10217.b`, failed using the standard version of *ENCGEN*. The runtime output indicated that all data records were flagged as: "`FDSC inconsistent with earth received time`" and thus no output was generated. An investigation, using the specialized 'no timecheck' version of *ENCGEN* revealed that part of DOY 216 and all records from DOY 217 were flagged and excluded.

The same result was found with the Voyager-2 final EDRs '`crsf10210.b`' and '`crsf10217.b`' processed with our standard production software. However, unlike the early problem, both of the final EDRs were processable with the 'no timecheck' version of *ENCGEN* . The encyclopedia files generated included all of the data from the standard programs plus the missing data. No data that should have been excluded was detected (Fig. 2). See **Appendix D** for the detailed emails and notes covering the DOY 216-217 issues.

Database Correction for Final EDR CRSF10176.b

The standard encyclopedia files with the missing data were replaced in the database with the encyclopedia files generated with the specialized 'no timecheck' software.

`crsf10210.b`	1177149 – 1177737	standard	Original Final Encyclopedia File
`crsf10217.b`	1177909 – 1178437	standard	Original Final Encyclopedia File
`crsf10210.b`	1177149 – 1177761	no timecheck	New Final Encyclopedia File
`crsf10217.b`	1177822 – 1177437	no timecheck	New Final Encyclopedia File

Final Explanation From JPL

Whenever the FDS clock is reset, the SCLK and the SCET will be disconnected from the existing time trend. Therefore, you should not use your "timechecks" for your processing for about 2 days until the SCLK and SCET sync up.

If we have any clock reset in the future, I'll make sure to advise you about the event so that you can stop your "timechecks" before your processing.

As noted in **Appendix D** JPL created several additional test EDRs to assist in analyzing the excluded data problem. These files were not helpful.

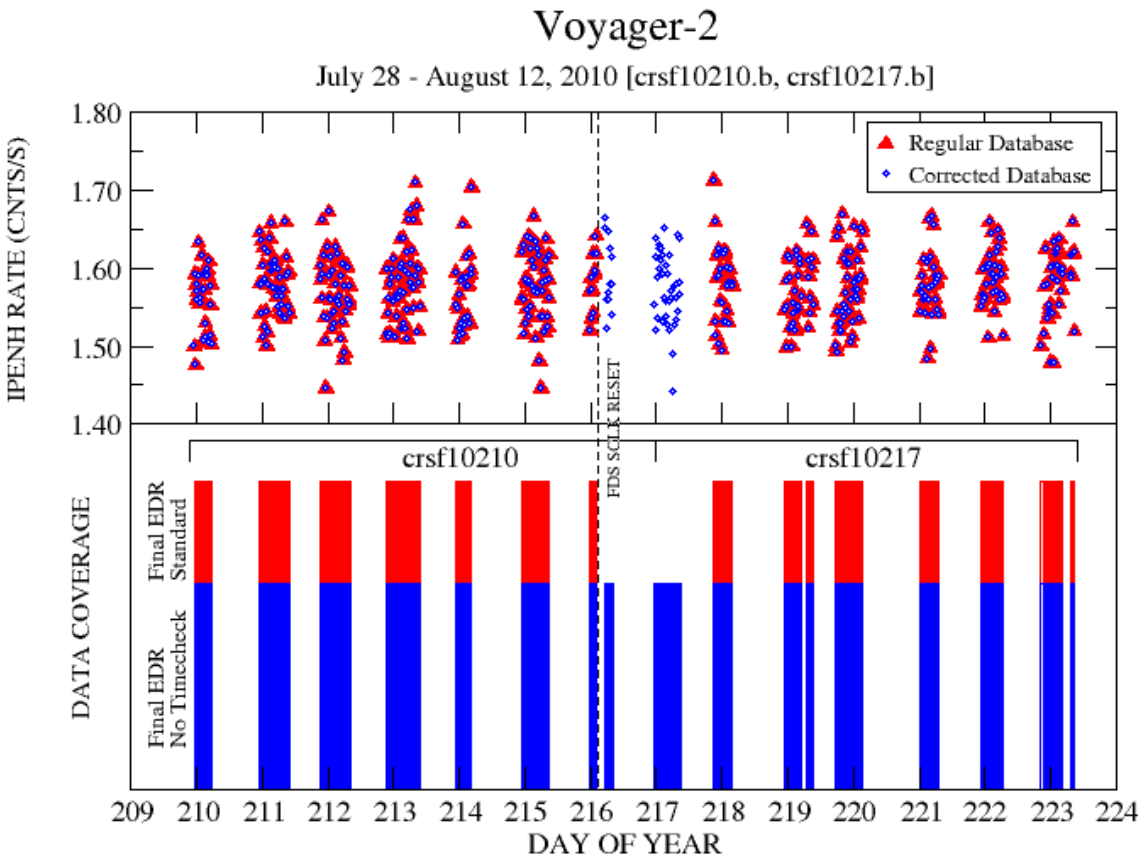


Figure 2 Top: Voyager-2 IPENH rate showing the data differences between the corrected and uncorrected data base. Bottom : Voyager-2 coverage representation . The red boxes represents the final EDRs processed with the standard *ENCGEN* program. The blue represents the final EDRs processed with the specialized *ENCGEN* program.

APPENDIX A

Email and Web Site Postings Covering the Voyager-2 Anomaly

The initial problem was reported to CRS by Regina Wong (JPL) on April 26:

Hi Everyone,

There's seems to be a problem on locking up the Voyager 2's CR-5 data at DSS-34, 43 and 45.Problem occurred since DOY 112 beginning at 13:35 ERT. We are still trouble shooting and no Voyager 2 data until further notice.

Regards,

Regina Wong
Regina.F.Wong@jpl.nasa.gov
Voyager Science & Data Management
(818) 393-1035

The following emails and website postings clarified the problem and the correction process:

From: "Massey, Ed B (7050)" <ed.b.massey@jpl.nasa.gov>

Date: Wed, 28 Apr 2010 15:53:41 -0700
Subject: Voyager 2 Anomaly
Thread-Topic: Voyager 2 Anomaly

Voyager 2 Anomaly Status Report 28 April 2010

On Thursday, 22 April 2010, The DSN was unable to frame-sync Voyager 2 telemetry. The downlink signal level, symbol signal-to-noise ratio (SNR) and the Maximum Likely Convolution Decoder SNR were as predicted. The previous pass on Wednesday, 21 April was nominal in all respects and ended as predicted.

Late Thursday, a Two-Way Non-Coherent (TWNC) OFF command was sent to the spacecraft to help us determine the best lock frequency for future uplinks. Late Friday, a command was uplinked to prevent the operation (NOP) of a MAGROL scheduled for today, 28 April at 9:13.
Because we were unable to process the telemetry data, there was no way to verify if the NOP command was received by the spacecraft. We would be able to tell if the MAGROL occurred by observing the downlink AGC.

In the meantime, the technical experts were consulted and the team was able to eliminate the ground systems, transmitter and RF systems as possible causes. The telemetry is formatted by the Flight Data Subsystem (FDS) which points to it as a possible source of the problem.

The plan going forward was to use the first command opportunity after the planned MAGROL and the end of the current command moratorium to send commands to the spacecraft to shift to the 40 bps engineering mode (EL40). This will allow the team to check status bits and determine whether the onboard sequence is still working. These commands are planned for Friday, 30 April. Subsequently, a switch to 1200 bps would be made so that a full FDS memory readout could be accomplished.

An analysis by Steve Howard of the data received at the station shows that telemetry frames are longer than expected by 80 bits with every fourth frame only 60 bits longer. He compared the 26 April data with the last data received on 21 April and found that the actual frame time was 48.46 seconds instead of the nominal 48.0 seconds (an increase by nearly a half a second per frame, which exactly explains the 75, on average, extra bits we are seeing in the data. He asked, and it was agreed that the station configure a parallel string to try to process the telemetry at 158.5 bps as well as the nominal 160 bps. During this morning's pass, 28 April, the station configured a second Downlink Channel Controller (DCC). They still saw the 80/60 extra bits per frame at the backup DCC. The DCC could frame sync on those 80 bits longer but not the fourth frames. At any rate, the AMMOS can not currently process these data.

During this morning's contact it was determined that the MAGROL did not occur. However, there are other possible reasons why it might not have occurred. So, to verify that it was a result of the NOP being executed and not by some other occurrence, we need to command the spacecraft to EL40.

Commands to turn TWNC OFF and Switch to EL40 will be radiated on Friday afternoon, 30 April, and results received on Saturday afternoon, 1 May. The mode change to EL40 exercise a different part of the Flight Data Subsystem telemetry code and removes the Reed Solomon Encoding portion of the telemetry. The next steps will depend on what we learn on Saturday.

Email from Ed Stone April 28:

FYI. The FDS creates the frame format, so this is likely a problem in the FDS memory. Depending on what is learned from the 40 bps engineering mode, the next step may be an FDS memory dump at 1200 bps. The round trip light time is over 25 hours, so this will take time.

Ed

Posted on the Voyager website at JPL:
May 06, 2010

Engineers have shifted NASA's Voyager 2 spacecraft into a mode that transmits only spacecraft health and status data while they diagnose an unexpected change in the pattern of returning data. Preliminary

engineering data received on May 1 show the spacecraft is basically healthy, and that the source of the issue is the flight data system, which is responsible for formatting the data to send back to Earth. The change in the data return pattern has prevented mission managers from decoding science data.

The first changes in the return of data packets from Voyager 2, which is near the edge of our solar system, appeared on April 22. Mission team members have been working to troubleshoot and resume the regular flow of science data. Because of a planned roll maneuver and moratorium on sending commands, engineers got their first chance to send commands to the spacecraft on April 30. It takes nearly 13 hours for signals to reach the spacecraft and nearly 13 hours for signals to come down to NASA's Deep Space Network on Earth.

Posted on the Voyager website at JPL:
Updated May 17, 2010 at 5:00 p.m. PDT

One flip of a bit in the memory of an onboard computer appears to have caused the change in the science data pattern returning from Voyager 2, engineers at NASA's Jet Propulsion Laboratory said Monday, May 17. A value in a single memory location was changed from a 0 to a 1.

On May 12, engineers received a full memory readout from the flight data system computer, which formats the data to send back to Earth. They isolated the one bit in the memory that had changed, and they recreated the effect on a computer at JPL. They found the effect agrees with data coming down from the spacecraft. They are planning to reset the bit to its normal state on Wednesday, May 19.

Posted on the Voyager website at JPL:
Updated May 20, 2010 at 6:00 p.m. PDT

Engineers have successfully corrected the memory on NASA's Voyager 2 spacecraft by resetting a computer bit that had flipped. Reset commands were beamed up to the spacecraft yesterday, Wed., May 19, and engineering data received today confirm that the reset was successful. The Voyager team will continue monitoring the engineering data, and if the bit remains reset, commands to switch to the science data mode will be beamed up to Voyager 2 on Sat., May 22. Receipt of science data would then resume on Sun., May 23.

From: "Massey, Ed B (7050)" <ed.b.massey@jpl.nasa.gov>
Voyager 2 Status, 20 May 2010

Commands to reset the flipped bit in the Flight Data System were uplinked on yesterday. Today we confirmed via memory readout and checksums that the reset worked. The checksum is again correct. We are still in EL-40 engineering mode. If the checksum holds until Saturday, 22 May, we will send commands to changed the telemetry to 160-bit (CR-5T) cruise mode with confirmation expected on Sunday 23 May.

EdBM

Email from Ed Stone May 23:

Great news- V2 science data is being received!

_____

From: Ludwig, Roger (317B)
Sent: Sunday, May 23, 2010 8:47 AM
To: Estabrook, Polly (3320); Taylor, Francis H (337H); Mohr, David L (317B)
Cc: Massey, Ed B (7050); Hall, Jefferson C (314C)
Subject: RE: VGR2 160 bps frame sync anomaly

Polly, Dave, Jim,

About 5 minutes ago we achieved frame synch on 160C science/engineering telemetry. Success!

Roger

Posted on the Voyager website at JPL:
Updated May 24, 2010 at 5:30 p.m. PDT

Engineers successfully reset a computer onboard Voyager 2 that caused an unexpected data pattern shift, and the spacecraft resumed sending properly formatted science data back to Earth on Sunday, May 23. Mission managers at NASA's Jet Propulsion Laboratory in Pasadena, Calif., had been operating the spacecraft in engineering mode since May 6. They took this action as they traced the source of the pattern shift to the flip of a single bit in the flight data system computer that packages data to transmit back to Earth.
In the next week, engineers will be checking the science data with Voyager team scientists to make sure instruments onboard the spacecraft are processing data correctly.

Email from Regina Wong (JPL) on May 24:

Hi Everyone,

On Wednesday (5/19/2010), commands to reset the Flight Data System(FDS) flipped bit on Voyager 2 were sent. On Saturday (5/22/2010), commands to change the telemetry mode from EL-40 to CR-5T (160 bps) Cruise Science data were sent. On Sunday (5/23/2010) at
15:41:27 ERT (day 143), DSS-43 was in lock on the 160 bps as expected.

You can retrieve your Voyager 2 data (daily file) for day 143 from our SHOCK workstation.

Data look good at my area. Please let me know what your status is (good or bad), and let me know if you have any problems with the files.

Thank you!

Regina Wong
Regina.F.Wong@jpl.nasa.gov
Voyager Science & Data Management
(818) 393-1035

APPENDIX B

Supporting Email and Notes regarding the Voyager-2 ERT and SCET time discrepancy

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Email from Regina Wong (JPL) on May 24:

Hi Everyone,

On Wednesday (5/19/2010), commands to reset the Flight Data System(FDS) flipped bit on Voyager 2 were sent. On Saturday (5/22/2010), commands to change the telemetry mode from EL-40 to CR-5T (160 bps) Cruise Science data were sent. On Sunday (5/23/2010) at
15:41:27 ERT (day 143), DSS-43 was in lock on the 160 bps as expected.

You can retrieve your Voyager 2 data (daily file) for day 143 from our SHOCK workstation.

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Thank you!

Regina Wong
Regina.F.Wong@jpl.nasa.gov
Voyager Science & Data Management
(818) 393-1035

Email to Regina Wong (JPL) on May 24:

Hi Regina,

My initial attempt to process the V2 data file failed. I asked Nand to take a look at the file. I hope to get back with you soon.

Bryant

README File in '/cosmicra/voyager/edr/test/v2ert2/may2410.d'

```
05/24/2010

Processing of Voyager-2 data file 'crs10143.b' which was the first
available daily EDR after month long data outage due to spacecraft
FDS problem. It will not process with either the ‘src.test’ or
‘src.mod’ versions of ‘encgen’.
```

```
./doedr crs10143.b
53 records processed.


./doencgen 2 crs10143.bgsfc
2
Library file to process: crs10143.bgsfc
exec /cosmicra/voyager/encgen/src.mod/encgen 2 crs10143.bgsfc
child process exited abnormally
while executing
"exec /cosmicra/voyager/encgen/src.mod/encgen 2 crs10143.bgsfc >
crs10143.bgsfc.output"
("eval" body line 1)
invoked from within
"eval "$command > ${libraryFile}.output""
(file "./doencgen" line 38)
```

Email to/from Nand May 24 - 25 regarding the processing failure.

Nand,

Voyager-2 data is flowing again but I could not process the EDR from JPL. ENCGEN reported 'no output' from the Library file. All of the relevant files are at:

'/cosmicra/voyager/edr/test/v2ert2/may2410.d'

Bryant

--
Bryant Heikkila
Voyager Interstellar Mission
Wyle Information Systems
bryant.heikkila@gmail.com

From Nand:

The Earth Received time appears to be in error. One way light time is about 12 hours 42 minutes, but the difference between spacecraft time and earth received time on the EDR is >15 hours. I have sent a message to Regina. I may create an EDR with ERT consitent with spacecraft time in order that we can process the EDR and look at our data while we wait for a response from JPL.

From Nand:

Please plot some data from /tmp/nand/V21170722.A1170755 which contains the day 143 data processed ignoring the time consistency issue.

Nand,

I have plotted the new data for PENH, 2-3 MeV H and 30-56 MeV/n He. In each figure the new data in RED is compared to daily averaged old data in BLUE (offset) to make sure the magnitude looks good. I think the data looks OK in all three cases.

Bryant

Email from Nand to Regina Wong (JPL) on May 24:

From: Lal, Nand (GSFC-6720) [mailto:nand.lal-1@nasa.gov]
Sent: Monday, May 24, 2010 6:45 PM
To: Wong, Regina F. (JPL-315H)[Caltech-JPL]
Subject: RE: Voyager 2 Data

SCET and ERT are inconsistent. (ERT apparently in error). Difference between ERT and SCET on the EDR for day 143 is 15 hr 1607 sec. One way light time is 12hr 42min.

Email from Regina Wong (JPL) to Nand on May 25:

Hi Dr.Lal,

The inconsistent of the SCET and ERT may be caused by the recent FDS(Flight Data System)bit flip anomaly. During the time of the anomaly, the FDS clock did not increment and there's an offset difference of about 2 hrs. and 28 min.

The OWLT of Voyager 2 is 12 hrs 42 min. and 31 sec. plus the offset difference of 2 hrs and 28 min. is about 15 hrs. 10 min. 31 sec. This is very close to your finding.

The Voyager team is still investigating the FDS anomaly and will decide what to do when we have further information.

Please let me know if you have any other questions/problems.

Thanks,
Regina Wong

Regina.F.Wong@jpl.nasa.gov
Voyager Science and Data Management
(818) 393-1035

Email from Regina Wong (JPL) on May 27:

Hi Everyone,

With Voyager 2 returning back to CR-5T (160 bps) Cruise Science Data mode on day 143 at 15:41:27 ERT, the following are the initial status of all instruments:

Instrument	Date	Status
CRS	5/24/2010	May Be (Concern about the time difference between ERT and SCET)
LECP	5/24/2010	May Be (Initial processing is fine but needs Dr. Decker to confirm, he's on travel)
MAG	5/24/2010	Good
UVS	5/24/2010	Instrument is off
PWS	5/24/2010	Good
PLS	5/27/2010	Good

There is a timing offset (time delay) of about 2 hrs. and 28 min. during the Flight Data System (FDS) bit flip anomaly. The FDS clock was slowing down and caused the difference between the ERT and SCET is more than the OWLT. The Voyager team is still investigating the FDS anomaly and will decide what to do with the FDS timing problem.

Please let me know if you have any questions of problems.

Thank you,

Regina Wong
Regina.F.Wong@jpl.nasa.gov
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(818) 393-1035

APPENDIX C

Supporting Email and Notes regarding the Voyager-2 EDRs for DOY 176-183

Email to Regina Wong (JPL) on June 29:

Hi Regina,

We were not able to process the V2 daily EDR for DOY 176. There are 78 records but they seem to be all overlapping data. Can you take a look at the file again? Thanks.

Bryant

Email from Regina Wong on June 30:

Hi Bryant,

I am on family sick leave since 6/21/2010. I'll forward your message to Ron Hungerford (my alternate during my absence) and have him look into that. I came this morning mainly to create the new SCLK_SCET files. Voyager 2 had a FDS SCLK reset on Sunday (day 179) take correct the clock offset problem (off by about 2 hrs). You may have questions on day 180 and 181's daily files because Ron Hungerford created those files before I created the new SCLK_SCET.32 file.

Regina

Email from Ron Hungerford (JPL) on July 1:

Bryant,

Regina Wong is out for family illness for awhile. I am temporarily filling in for Voyager science questions this week. We examined the Day 176 source data:

S/C 32 DOY 176
DSS 45 Track

RAIN
DR C107455

Poor data
12:00:00 - 21:02:29 2 min 29 sec
12:14:29 - 23:21:41 2 min 12 sec

12:29:41 - 12:58:29 28 min 48 sec
13:13:41 - 13:16:53 3 min 12 sec
13:44:53 - 13:53:41 8 min 48 sec

Data received between will be corrupted… likely bad FDSC & SCET

Unfortunately, the rain over the station corrupted the telemetry stream, generating bad FDSCs. The bad FDSCs are translated to bad SCETs.

We have no tools to repair the data, so this is the best we can do.

Sorry,

Ron Hungerford
VGR Science Data Team

Email from Regina Wong (JPL) on June 30:

Hi Everyone,

I am off since 6/21/2010 due to family illness. Ron Hungerford is my alternate for the Voyager task.

Ron is doing the daily task and he created the final EDRs last week. The files are available for your access on our SHOCK workstation.

Voyager 2 had a FDS SCLK reset on day 179 with the following information:

2010-179/16:25:57.564 (ERT) = 2010-179/03:42:57.465 (SCET)

I created a new file (for JPL's central data system) to reflect this reset this morning and they cannot put the new correlation file until the DSS pass is over for day 181. Your data for days 180 and 181 will still have the old clock issue until the new file is on the system starting on day 182. If you have any questions on the data, please contact Ron Hungerford, any other questions about Voyager's science activities, please contact Jefferson Hall.

I'll let everyone know when I'll be back to work.

Thanks,
Regina Wong
Voyager Science and Data Management
(818) 393-1035

07/08/2010

```
The autoprocessing of 'crsf10175.b' using the 'no timecheck' version
of ENCGEN ("/tmp/nand/test/src/encgen") generated no output.
Proceeding to handprocessing.

/cosmicra/voyager/edr/test/v2ert2> ./doedr crsf10175.b
646 records processed.

/cosmicra/voyager/edr/test/v2ert2> ./doencgen 2 crsf10175.bgsfc
2
Library file to process: crsf10175.bgsfc
exec /cosmicra/voyager/encgen/src.test/encgen 2 crsf10175.bgsfc

Runtime error messages show:
Overlapping data at record XX from records 14 - 506


Reprocessed using the standard
"/cosmicra/voyager/encgen/src.test/encgen"
program. This run successfully generated an Encyclopedia file.
Runtime error messages indicate:

From record ~285 - ~363 the "FDSC inconsistent with earth received
time."
```

Email from Regina Wong (JPL) on July 20:

Hi Everyone,

I am back to work.

Ron Hungerford generated some new final EDRs for both spacecraft during my absence. The files are from day 176 to 198 and available for you to retrieve from our SHOCK workstation.

Regards,
Regina Wong
Voyager Science and Data Management

APPENDIX D

Supporting Email and Notes regarding the Voyager-2 EDRs for DOY 217

Email from Regina Wong (JPL) on August 9:

Hi Everyone,

Voyager 2 had another FDS SCLK Reset at 2010-216/15:42 ERT. The purpose of this reset was to have the CCS clock and the FDS clock on the Voyager 2 to sync correctly in order to do sequence in the future. The latest final EDRs which I just posted includes the new clock adjustment.

Please let me know if you have any problems.

Regina Wong
Regina.F.Wong@jpl.nasa.gov
Voyager Science and Data Management
(818) 393-1035

Email from Regina Wong (JPL) on August 9:

Hi Bryant,

There's a new SCLK_SCET.32 file for Voyager 2 for you to access from our SHOCK workstation.

There's a FDS clock reset at :

201-216/15:42 (ERT)

This new SCLK_SCET file is for that adjustment.

Regina

Email to Regina Wong (JPL) and Nand Lal (GSFC) on August 13:

Hi Regina,

CRS was unable to process the V2 daily for DOY 217. However, all subsequent V2 daily files have processed correctly.

Thanks.

Bryant

Email from Regina Wong (JPL) on August 13:

Hi Bryant,

Can you please explain a little more about V2's DOY 217 daily file? What kind of problem you have with it? What would you like me to do? Is it the entire day or just at certain time?

Regina

Email to Regina Wong (JPL) on August 13:

Hi Regina,

For DOY 217 all 62 data records get flagged by our software as: "FDSC inconsistent with earth received time." This is the same problem we had before the first FDS clock reset. As long as we can process the final EDRs for that day then we will be OK.

Bryant

Email from Regina Wong (JPL) on August 13:

Hi Bryant,

I just sent out a message about the new final EDRs that were created today. Please process V2's file and see if you still have problems with the data for DOY 217.

Thanks,

Regina

Email to Nand on August 13:

Nand,

I will process the final EDRs this weekend and see what happens with DOY 217 V2 data. The daily file would not process due to all of the records had time errors.

If the finals process OK then no we do not need the old program.

Bryant

Sent from my iPad

Email to Regina Wong (JPL) on August 19:

Hi Regina,

After processing the final EDRS for Voyager-2 (crsf10210.b, crsf10217.b) we are missing data for part of day 216 and all of day 217 unless we turn off our time checks. Since this occurs right after the FDS Clock Reset and at no other time covered in these EDRs we think something unusual must have happened with respect to the reset. If you have time can you look into this for us. The attached figure shows the results of our processing with the red sections showing standard processed data (y-axis is not important). The blue sections show data processed with our time checks turned off. Our processing shows that all 62 data records of V2 DOY 217 are flagged with incorrect times as well as part of DOY 216. Thanks.

Bryant

Email from Regina Wong (JPL) on August 20:

Hi Bryant,

I have some questions about this problem that you have:

1. What is “time checks”? What does it do? Is it a piece of software at your local area?
2. If you turn off your “time checks”, does it mean that you get all of the data?
3. When we had the first FDS Clock Reset for Voyager 2 on day 179 at 16:26 (ERT), did you have this kind of problem?
4. Did you have problem with part of DOY 216 and all of DOY 217 only? Starting on DOY 218, everything is fine?
5. The 62 data records of V2 for part of DOY 216 and DOY 217 are flagged with incorrect times, what does that mean? What times did you expect to see and what times did you actually have?

Hope you can help me out with these questions so that I can investigate the problem.

Thanks,
Regina

Email to Regina Wong (JPL) on August 23:

Hi Regina,

I have added more information following each of your questions below.

Bryant

On Fri, Aug 20, 2010 at 6:49 PM, Wong, Regina F (315H) <regina.f.wong@jpl.nasa.gov> wrote:

Hi Bryant,

I have some questions about this problem that you have:

1. What is “time checks”? What does it do? Is it a piece of software at your local area?

Yes, "time checks" is a general term used to describe data quality analysis with respect to the times during our processing of the EDRs.

2. If you turn off your “time checks”, does it mean that you get all of the data?

Yes. That is what I tried to show you with the previously sent figure. If we turn off our "time checks" then we get all of the data. With our "time checks" running part of DOY 216 and all of DOY 218 fail and are excluded. The failed records start right after the reset.

3. When we had the first FDS Clock Reset for Voyager 2 on day 179 at 16:26 (ERT), did you have this kind of problem?

We had the problem until you reset the FDS clock but not after. I am going to reprocess some files to make sure we did not miss some data during the reset. I will get back with you on this one.

4. Did you have problem with part of DOY 216 and all of DOY 217 only? Starting on DOY 218, everything is fine?

Yes, the only problem is with part of DOY 216 and DOY 217. DOY 218 is fine. We believe it has something to do with the reset. Did it fail or get sent wrong the first time and they had to resend the command?

5. The 62 data records of V2 for part of DOY 216 and DOY 217 are flagged with incorrect times, what does that mean? What times did you expect to see and what times did you actually have?

Our "time checks" compares the expected FDSC derived from the ERT to the actual value in the EDR. The SCET is computed from the ERT which is assumed to be correct. If these checks fail then the record(s) are flagged. Below is one of the flagged records showing a diffrerence of about 13 minutes between the computed SCET and the SCET (old) in the EDR. The numbers are in milliseconds.

Replacing old SCET 1060037763327.000000 with 1060036995883.315796

Email to Regina Wong (JPL) on August 23:

Hi Regina,

Thank you for asking question number 3 below. After looking into it some more I discovered that we did indeed miss data. In this case part of DOY 179 and all of DOY 180. The data is there of course but it fails our time checks. I have attached another figure showing the missing data. Ignore the green data sets, that is another mystery for Nand and me to solve.

Bryant

Email from Regina Wong (JPL) on August 25:

Hi Bryant,

I generated two new final files for you to retrieve. Information about the files are as follow:

	File Name	Start Time	Stop Time
1)	Crsf10179.b	2010-179/17:27 (ERT)	2010-180/19:50 (ERT)
2)	Crsf10216.b	2010-216/17:43 (ERT)	2010-217/21:30 (ERT)

Please let me know if you still have problems with the data on these files as soon as you can.

Thank you,

Regina Wong

Email to Regina Wong (JPL) and Nand Lal (GSFC) on August 26:

Regina,

I was unable to process these files. Both EDRs failed to produce output with our standard production software or our special version without timechecks. These files are really big. They are 3 times the size of a final EDR but only contain about a days data based on the times in your email. In addition, there are over 900 records in each file but I would expect less than 100. The error messages indicate every record consists of overlapping data.

Bryant

Email from Regina Wong (JPL) on August 31:

Hi Bryant,

I regenerated two new files for you to retrieve from our SHOCK workstation. File names, start and stop time are same as before.
Please let me know if these files are acceptable.

Thank you,
Regina Wong

Email to Regina Wong (JPL) on September 2:

Hi Regina,

I was not able to process these files using our standard programs but I was able to process them using our special version without timechecks. Using a combination of daily and final EDRs I have been able to fill our data base so that we do not have missing data. I don't know if you want to continue to investigate this or simply note that there is a problem with the times right after a FDS Clock reset.
I have a couple more quality checks to run and then discuss the results with Nand, but I don't want to hold up the final EDR generation of new data.

Thanks,
Bryant

Email from Regina Wong (JPL) on September 3:

Hi Bryant,

Whenever the FDS clock is reset, the SCLK and the SCET will be disconnected from the existing time trend. Therefore, you should not use your “timechecks” for your processing for about 2 days until the

SCLK and SCET sync up. The following indicates the time equivalent on the start and stop time for the two files which I generated for you on 8/31/2010:

1)	Crsf10179.b	Start Time	2010-179/17:27 (ERT)	= 43140.02.633 (SCLK)
				= 2010-179/04:43:59.783 (SCET)
		Stop Time	2010-180/19:50 (ERT)	= 43173.01.384 (SCLK)
				= 2010-180/07:06:56.655 (SCET)
2)	Crsf10216.b	Start Time	2010-216/17:43 (ERT)	= 44253.33.063 (SCLK)
				= 2010-216/04:56:55.184 (SCET)
		Stop Time	2010-217/21:30 (ERT)	= 44288.16.527 (SCLK)
				= 2010-217/08:43:46.831 (SCET)

These events are not normal mission activities; we just wanted to adjust the recent Voyager 2's FDS memory problem.

If we have any clock reset in the future, I'll make sure to advice you about the event so that you can stop your "timechecks" before your processing.

Please let me know if you still have any questions or problems.

Regina Wong